The math you need this week

Fractions and percentages · the sum sign Σ · the exponential e^x · areas of rectangles · about 10 minutes

Before you start

This is a **review**, not new material. Week 3 builds probability from these four small tools. Read this file, then do the **Week 3 Math check** on Moodle (no grade, as many tries as you like). If a question goes wrong, the feedback tells you which section to read again.

M3.1 Fractions, decimals and percentages

A fraction "part over whole" can be written three ways: $\frac{56}{60} = 0.933 = 93.3\%$. To get a percentage, multiply the decimal by 100. To go back, divide by 100.

Worked example

In a catch of 100 fish, 59 are light: $59/100 = 0.59 = 59\%$. Of the 60 sea bass, 56 are light: $56/60 = 0.933$.
Be careful: the whole matters. "56 of the 60 sea bass" (0.933) is not "56 of the 59 light fish" (0.949).

In NumPy

```
56 / 60          # 0.9333
round(56 / 60 * 100, 1)   # 93.3 (percent)
```

M3.2 The sum sign Σ

$\sum_x f(x)$ means "add $f(x)$ for every value of x". It is a short way to write a long addition.

Worked example

For $x = 0, 1, 2, 3$: $\sum_x x = 0 + 1 + 2 + 3 = 6$.
With weights $w = (0.5, 0.3, 0.15, 0.05)$: $\sum_x x\, w_x = 0(0.5) + 1(0.3) + 2(0.15) + 3(0.05) = 0.75$.

In NumPy

```
x = np.array([0, 1, 2, 3]);  w = np.array([0.5, 0.3, 0.15, 0.05])
np.sum(x * w)    # 0.75
np.sum(w)        # 1.0
```

M3.3 The exponential e^x

$e \approx 2.718$ is a fixed number, like π. The function e^x, written `exp(x)`, appears in the normal density as $e^{-z^2/2}$. You only need to *evaluate* it with a calculator or NumPy.

- $e^0 = 1$. For negative powers $e^{-a} = 1/e^a$ is between 0 and 1, and gets smaller as a grows.
- $e^{-z^2/2}$ is largest (1) at $z = 0$ and falls fast when z moves away from 0.

Worked example

$z = 1$: $e^{-1/2} = e^{-0.5} = 0.6065$. $z = 2$: $e^{-4/2} = e^{-2} = 0.1353$. $z = 4$: $e^{-16/2} = e^{-8} = 0.000335$.
Be careful with the order: first square z, then halve, then put a minus sign, then take e to that power.

In NumPy

```
np.exp(-0.5)     # 0.6065
z = 2;  np.exp(-z**2 / 2)   # 0.1353
```

M3.4 Area of a rectangle, and area under a curve

The area of a rectangle is width $\times$ height. A histogram is a row of rectangles, so its total area is the sum of width $\times$ height over all bars. The area under a smooth curve is the same idea with very thin bars.

Worked example

A bar from 190 to 195 has width 5 and height 0.0363: area $5 \times 0.0363 = 0.181$.
Three bars of width 2 with heights 0.1, 0.2 and 0.2 have total area $2(0.1 + 0.2 + 0.2) = 1.0$.

In NumPy

```
heights = np.array([0.1, 0.2, 0.2]);  width = 2
np.sum(heights * width)    # 1.0
```